

tral processing units (for desktops) that had an in-built refrigeration system to bring the system temperature down to -36°C, wherein the best performance was achieved.

It is only when temperatures drop beyond -230°C, such as in space applications, that the problem begins, as the properties of silicon start changing at that point. Since the applications are very niche, the problem is somehow overcome using insulation and heating systems, or special packaging and materials.

The bigger problem—with greater applications—that silicon makers are more concerned about today is extreme heat. We know how heat affects a laptop. In summers, we often end up having blowers or laptop coolers to bring down the temperature to avoid performance degradation. This is at temperatures in the range of 40°C-50°C. Imagine how temperatures in the range of 125°C or higher would affect electronic components!

If conventional electronics is to be used at extreme temperatures, it would require passive or active cooling, using refrigerators and heat sinks. Probes used for well-logging, for example, may be placed in vacuum-insulated vessels to protect them from the hot environment. In addition, the electronics may be thermally connected to a heat sink, which uses materials like water or bismuth alloy that can absorb large amount of heat without a substantial temperature increase. This is usually done by employing the material's phase change from solid to liquid, which absorbs a large amount of heat (the latent heat of fusion).

However, all such cooling techniques—both passive and active—have problems. Passive techniques have a limited lifetime, while active techniques require additional power and sub-systems. Active techniques may also be too disturbing to the environment because of the additional heat that is ejected from the system. All the techniques add weight, size and some degree of complexity—which is unacceptable because extreme-temperature

Tough weather, mate!

Electronic systems are often required to work in extremely harsh temperature conditions, beyond industrial or even military ranges. Here are some examples.

Down-hole oil and gas industry. This is one of the largest markets for extreme-temperature electronics. Well-loggers or probes sent into geothermal wells have sensors and signal processing equipment, which have to withstand temperatures as high as 300°C. The typical geothermal gradient is 25°C/km as you go closer to the centre of the Earth. So the deeper the underground well, the worse the operating temperatures.

Avionics. The distributed control systems used in modern aircrafts require engine control systems to be closer to the engine, to reduce complexity of interconnections and increase reliability. Since the electronic components are closer to the engine, these now have to sustain much higher temperatures.

Automotive. The electromechanical or mechatronic systems in modern automobiles require sensors, signal conditioning and control electronics to be located close to engines and heat sources. Electric and hybrid vehicles also require power electronics with high-energy density for converters, motor controls and charging circuits that are also associated with high temperatures.

Space exploration. A space probe that goes to Neptune should be able to operate at -230°C. On Venus, the temperature would be close to 500°C. When orbiting the Earth, the temperature might drop to -230°C near the lunar poles. Beyond tolerating such temperatures, the electronics should also be able to handle sudden or gradual changes in temperature along the path to its destination. Further, this complex industry requires not just temperature resistance but also radiation-hardened components that can withstand particle radiation and high-energy electromagnetic radiation.

Astrophysical equipment. Astrophysical equipment, such as bolometers and infrared space telescopes, require electronics that can withstand very low temperatures.

Medical equipment. Medical equipment that handle liquid nitrogen and other such compounds often need to handle very cold temperatures.

Surveillance. Sensors and signal processors used in military surveillance equipment must withstand very harsh weather conditions as they may be deployed anywhere.

LED lighting. LED lighting, especially centrally-managed lighting systems, require control systems, signal processors, etc that can withstand high temperatures generated by the light.

Other applications including undersea cabling and industrial use.

applications are generally critical and space constrained.

Even if sufficient ventilation and cooling is incorporated into the system, electronic components still need to be extensively tested and validated using lab ovens, etc, to confirm whether they can withstand such temperatures. The component maker generally would not give any assurance or guarantee for the functioning of standard components at extreme temperatures; so deployment is at the user's own risk. Plus, even if the components per se work, sometimes the packaging might not be tolerant to very high temperatures, in which case the die has to be purchased and repackaged for the application. And, despite all this effort, there are chances that the component could fail due to other factors like vibration, interference or electro-migration in metal interconnects, which are very difficult to detect during testing.

All this gives rise to the need for special electronics that can withstand extreme temperatures, especially on the higher end.

Modern technologies

New materials, testing, qualification and packaging techniques have helped silicon makers to come up with customised silicon chains that can work in extreme temperatures with high reliability.

"Integrated circuit (IC) technology from companies like ADI has produced devices that can operate reliably at elevated temperatures with guaranteed datasheet specifications. Advances have been made in process technology, circuit design and layout techniques. Managing many key device characteristics is crucial for successful, high-performance operation at elevated temperatures. One of the most important and well-known challenges is posed by the increased